hdoc day 49 by trishil chauhan

Problem Statement

Read a positive integer n and pass its address to **pivot_integer()**. Find an integer x from 1 to n for which the inclusive sum from 1 to x equals the inclusive sum from x to n. Print x, or -1 if no such integer exists.

C Program

```
#include <stdio.h>

void pivot_integer(const int *n) {
    long long total = (long long)(*n) * (*n + 1LL) / 2;
    long long leftSum = 0;

    for (int x = 1; x <= *n; x++) {
        leftSum += x;
        long long rightSum = total - leftSum + x;

        if (leftSum == rightSum) {
            printf("Pivot integer: %d\n", x);
            return;
        }
    }
    printf("Pivot integer: -1\n");
}

int main(void) {
    int n;
    printf("Enter a positive integer n: ");
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("Invalid input. Enter a positive integer.\n");
        return 1;
    }
    pivot_integer(&n);
    return 0;
}
```

Sample Output

```
Example 1:
Enter a positive integer n: 8
Pivot integer: 6

Example 2:
Enter a positive integer n: 4
Pivot integer: -1
```

Explanation

For n = 8 and x = 6, the left sum is 1 + 2 + 3 + 4 + 5 + 6 = 21, and the right sum is 6 + 7 + 8 = 21. The pivot value is included in both sums. The function receives a pointer to n, uses *n to read it, and

checks each candidate from 1 to n.

Complexity: O(n) time and O(1) extra space.